Full Roadmap Tracker — Java + Spring Boot + HLD/System Design

Span: Mon, Sep 14, 2026 → Mon, Feb 01, 2027 | Off: Sat & Sun | Hackathon gaps: Sep 25–27 & Oct 16–19, 2026

Har phase ke start pe resource link diya hai, aur har row mein completion ke waqt jo git commit karna hai wo already likha hua hai — bas us din ka kaam khatam karke wahi commit message use kar dena.

PHASE 0 — Java Full Course (CoderArmy, 57 videos)

Pace: 2 videos/day | 29 working days | Start: Mon Sep 14, 2026 → End: Tue Oct 27, 2026

Resource: https://www.youtube.com/playlist?list=PLQEaRBV9gAFsR15tNo2QLF9d2qc-c018p

(Note: source doc mein sirf HLD phase ke commits explicitly diye the. Java/Spring Boot ke liye same convention pe maine commit names bana diye hain — agar tum koi aur naming chahte ho to bata dena.)

Day	Date	Videos	Commit (on completion)
1	Mon, Sep 14, 2026	1–2	java-day-01-videos-1-2
2	Tue, Sep 15, 2026	3–4	java-day-02-videos-3-4
3	Wed, Sep 16, 2026	5–6	java-day-03-videos-5-6
4	Thu, Sep 17, 2026	7–8	java-day-04-videos-7-8
5	Fri, Sep 18, 2026	9–10	java-day-05-videos-9-10
—	Sat–Sun, Sep 19–20	OFF	—
6	Mon, Sep 21, 2026	11–12	java-day-06-videos-11-12
7	Tue, Sep 22, 2026	13–14	java-day-07-videos-13-14
8	Wed, Sep 23, 2026	15–16	java-day-08-videos-15-16

Day	Date	Videos	Commit (on completion)
9	Thu, Sep 24, 2026	17–18	java-day-09-videos-17-18
—	Fri–Sun, Sep 25–27	HACKATHON (skip)	—
10	Mon, Sep 28, 2026	19–20	java-day-10-videos-19-20
11	Tue, Sep 29, 2026	21–22	java-day-11-videos-21-22
12	Wed, Sep 30, 2026	23–24	java-day-12-videos-23-24
13	Thu, Oct 1, 2026	25–26	java-day-13-videos-25-26
14	Fri, Oct 2, 2026	27–28	java-day-14-videos-27-28
—	Sat–Sun, Oct 3–4	OFF	—
15	Mon, Oct 5, 2026	29–30	java-day-15-videos-29-30
16	Tue, Oct 6, 2026	31–32	java-day-16-videos-31-32
17	Wed, Oct 7, 2026	33–34	java-day-17-videos-33-34
18	Thu, Oct 8, 2026	35–36	java-day-18-videos-35-36
19	Fri, Oct 9, 2026	37–38	java-day-19-videos-37-38
—	Sat–Sun, Oct 10–11	OFF	—
20	Mon, Oct 12, 2026	39–40	java-day-20-videos-39-40
21	Tue, Oct 13, 2026	41–42	java-day-21-videos-41-42
22	Wed, Oct 14, 2026	43–44	java-day-22-videos-43-44
23	Thu, Oct 15, 2026	45–46	java-day-23-videos-45-46
—	Fri–Mon, Oct 16–19	HACKATHON (skip)	—
24	Tue, Oct 20, 2026	47–48	java-day-24-videos-47-48
25	Wed, Oct 21, 2026	49–50	java-day-25-videos-49-50

Day	Date	Videos	Commit (on completion)
26	Thu, Oct 22, 2026	51–52	java-day-26-videos-51-52
27	Fri, Oct 23, 2026	53–54	java-day-27-videos-53-54
—	Sat–Sun, Oct 24–25	OFF	—
28	Mon, Oct 26, 2026	55–56	java-day-28-videos-55-56
29	Tue, Oct 27, 2026	57 (last video)	java-day-29-video-57-final

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PHASE 0.5 — Spring Boot Full Course (CoderArmy, 41 videos)

Pace: 2 videos/day | 21 working days | Start: Wed Oct 28, 2026 → End: Wed Nov 25, 2026

Resource: https://www.youtube.com/playlist?list=PLEYgx5hMdopw *(source doc mein ye playlist ID incomplete lag rahi thi — link confirm/verify kar lena before starting.)*

Day	Date	Videos	Commit (on completion)
1	Wed, Oct 28, 2026	1–2	spring-day-01-videos-1-2
2	Thu, Oct 29, 2026	3–4	spring-day-02-videos-3-4
3	Fri, Oct 30, 2026	5–6	spring-day-03-videos-5-6
—	Sat–Sun, Oct 31–Nov 1	OFF	—
4	Mon, Nov 2, 2026	7–8	spring-day-04-videos-7-8
5	Tue, Nov 3, 2026	9–10	spring-day-05-videos-9-10
6	Wed, Nov 4, 2026	11–12	spring-day-06-videos-11-12
7	Thu, Nov 5, 2026	13–14	spring-day-07-videos-13-14
8	Fri, Nov 6, 2026	15–16	spring-day-08-videos-15-16

Day	Date	Videos	Commit (on completion)
—	Sat–Sun, Nov 7–8	OFF	—
9	Mon, Nov 9, 2026	17–18	spring-day-09-videos-17-18
10	Tue, Nov 10, 2026	19–20	spring-day-10-videos-19-20
11	Wed, Nov 11, 2026	21–22	spring-day-11-videos-21-22
12	Thu, Nov 12, 2026	23–24	spring-day-12-videos-23-24
13	Fri, Nov 13, 2026	25–26	spring-day-13-videos-25-26
—	Sat–Sun, Nov 14–15	OFF	—
14	Mon, Nov 16, 2026	27–28	spring-day-14-videos-27-28
15	Tue, Nov 17, 2026	29–30	spring-day-15-videos-29-30
16	Wed, Nov 18, 2026	31–32	spring-day-16-videos-31-32
17	Thu, Nov 19, 2026	33–34	spring-day-17-videos-33-34
18	Fri, Nov 20, 2026	35–36	spring-day-18-videos-35-36
—	Sat–Sun, Nov 21–22	OFF	—
19	Mon, Nov 23, 2026	37–38	spring-day-19-videos-37-38
20	Tue, Nov 24, 2026	39–40	spring-day-20-videos-39-40
21	Wed, Nov 25, 2026	41 (last video)	spring-day-21-video-41-final

(Spring Boot khatam hote hi agle working day — Thu Nov 26 — se seedha HLD start.)

PHASE 1 — High Level Design (HLD) — 2 Hours/Day

48 working days | Start: Thu Nov 26, 2026 → End: Mon Feb 01, 2027 | One system per day rule

Primary resources (all weeks):

- donnemartin/system-design-primer — https://github.com/donnemartin/system-design-primer
- ashishps1/awesome-system-design-resources — https://github.com/ashishps1/awesome-system-design-resources
- Full 15-repo list is in the Reference Section at the end of this file.

Week 1 — Foundations + Core Concepts (Target: Distributed Session Store)

Resources: system-design-primer, alex/what-happens-when — https://github.com/alex/what-happens-when

Day	Date	Topic	Commit (on completion)
1	Thu, Nov 26, 2026	Networking Foundations	hld-day-01-networking-foundations
2	Fri, Nov 27, 2026	Scalability/Availability/Reliability/SPOF	hld-day-02-scalability-availability
3	Mon, Nov 30, 2026	CAP Theorem + PACELC	hld-day-03-cap-pacelc
4	Tue, Dec 01, 2026	Session Store: Consistency + Availability	hld-day-04-session-store-consistency-availability
5	Wed, Dec 02, 2026	Session Store: Hashing + Partition Challenge	hld-day-05-session-store-hashing-partition
6	Thu, Dec 03, 2026	Session Store: Fault Tolerance + Deliverable	hld-day-06-session-store-deliverable

Week 2 — Networking & API Layer (Target: Ride-Hailing App API)

Resources: system-design-primer, ashishps1, Chalarangelo/30-seconds-of-code — https://github.com/Chalarangelo/30-seconds-of-code, CodingChallengesFYI/SharedSolutions — https://github.com/CodingChallengesFYI/SharedSolutions

Day	Date	Topic	Commit (on completion)
7	Fri, Dec 04, 2026	DNS + Load Balancing	hld-day-07-dns-load-balancing
8	Mon, Dec 07, 2026	Reverse Proxy + API Gateway	hld-day-08-reverse-proxy-gateway
9	Tue, Dec 08, 2026	CDN	hld-day-09-cdn
10	Wed, Dec 09, 2026	Ride API: Contract + Style Decision	hld-day-10-ride-api-contract
11	Thu, Dec 10, 2026	Ride API: Real-Time Tracking	hld-day-11-ride-api-realtime
12	Fri, Dec 11, 2026	Ride API: Idempotency + Rate Limiter + Deliverable	hld-day-12-ride-api-deliverable

Week 3 — Database Layer (Target: Ride-Tracking App DB)

Resources: system-design-primer, ashishps1, awesome-scalability — https://github.com/binhnguyennus/awesome-scalability

Day	Date	Topic	Commit (on completion)
13	Mon, Dec 14, 2026	ACID + SQL vs NoSQL	hld-day-13-acid-sql-nosql
14	Tue, Dec 15, 2026	Database Indexes + Bloom Filters	hld-day-14-indexes-bloom-filters
15	Wed, Dec 16, 2026	Replication	hld-day-15-replication
16	Thu, Dec 17, 2026	Ride DB: Sharding + Federation	hld-day-16-ride-db-sharding-federation
17	Fri, Dec 18, 2026	Ride DB: Denormalization + Tuning Challenge	hld-day-17-ride-db-denormalization-tuning

Day	Date	Topic	Commit (on completion)
18	Mon, Dec 21, 2026	Ride DB: Architecture + Scaling Plan + Deliverable	hld-day-18-ride-db-deliverable

Week 4 — Caching Layer (Target: Product Catalog Caching Layer)

Resources: ashishps1, InterviewReady/system-design-resources — https://github.com/InterviewReady/system-design-resources, awesome-scalability, CodingChallengesFYI/SharedSolutions

Day	Date	Topic	Commit (on completion)
19	Tue, Dec 22, 2026	Caching 101 + Strategies	hld-day-19-caching-strategies
20	Wed, Dec 23, 2026	Cache Eviction Policies	hld-day-20-cache-eviction
21	Thu, Dec 24, 2026	Distributed Caching	hld-day-21-distributed-caching
22	Fri, Dec 25, 2026	Catalog: Cache Strategy + CDN Split	hld-day-22-catalog-cache-strategy
23	Mon, Dec 28, 2026	Catalog: Case Study + Build Challenge	hld-day-23-catalog-case-study-build
24	Tue, Dec 29, 2026	Catalog: Invalidation + Stampede + Deliverable	hld-day-24-catalog-caching-deliverable

Week 5 — Async, Messaging & Distributed Systems (Target: Order-Processing Async Pipeline)

Resources: ashishps1, InterviewReady/system-design-resources, arpit20adlakha/Computer-Science-Papers-For-System-Design — https://github.com/arpit20adlakha/Computer-Science-Papers-For-System-Design

Day	Date	Topic	Commit (on completion)
25	Wed, Dec 30, 2026	Message Queues + Pub/Sub	hld-day-25-mq-pubsub
26	Thu, Dec 31, 2026	Change Data Capture + Event-Driven Basics	hld-day-26-cdc-event-driven
27	Fri, Jan 01, 2027	Heartbeats + Service Discovery	hld-day-27-heartbeats-service-discovery
28	Mon, Jan 04, 2027	Order Pipeline: Queue Structure + Leader Election	hld-day-28-order-pipeline-queue-consensus
29	Tue, Jan 05, 2027	Order Pipeline: Locking + Consensus Challenge	hld-day-29-order-pipeline-locking-consensus
30	Wed, Jan 06, 2027	Order Pipeline: Gossip + Circuit Breaker + Deliverable	hld-day-30-order-pipeline-deliverable

Week 6 — Architecture Patterns + Tradeoffs (Target: any 1 sample system)

Resources: ashishps1, addyosmani/essential-js-design-patterns — https://github.com/addyosmani/essential-js-design-patterns

Day	Date	Topic	Commit (on completion)
31	Thu, Jan 07, 2027	Client-Server + Microservices Architecture	hld-day-31-client-server-microservices
32	Fri, Jan 08, 2027	Serverless + Event-Driven Architecture	hld-day-32-serverless-event-driven
33	Mon, Jan 11, 2027	P2P Architecture + Pattern Selection	hld-day-33-p2p-pattern-selection
34	Tue, Jan 12, 2027	Tradeoffs Part 1	hld-day-34-tradeoffs-1
35	Wed, Jan 13, 2027	Tradeoffs Part 2	hld-day-35-tradeoffs-2

Day	Date	Topic	Commit (on completion)
36	Thu, Jan 14, 2027	Tradeoffs Part 3 + Deliverable	hld-day-36-tradeoffs-deliverable

Week 7 — Easy Interview Problems, Hands-On (full interview format daily)

Resources: system-design-primer, ashishps1, 30-seconds-of-code, InterviewReady/system-design-resources

Day	Date	Topic	Commit (on completion)
37	Fri, Jan 15, 2027	Design URL Shortener (TinyURL)	hld-day-37-url-shortener
38	Mon, Jan 18, 2027	Design Rate Limiter (full interview format)	hld-day-38-rate-limiter-full-design
39	Tue, Jan 19, 2027	Design Parking Garage	hld-day-39-parking-garage-scale
40	Wed, Jan 20, 2027	Design Distributed Key-Value Store / Cache	hld-day-40-distributed-kv-store
41	Thu, Jan 21, 2027	Design Authentication System	hld-day-41-authentication-system
42	Fri, Jan 22, 2027	Design Web Crawler (Review Day)	hld-day-42-web-crawler-review

Week 8 — Medium + Hard, Capstone

Resources: ashishps1 Medium/Hard list, system-design-primer, design-guru/grokking-system-design (verify exact repo URL yourself, source doc naming was slightly inconsistent), yangshun/tech-interview-handbook — https://github.com/yangshun/tech-interview-handbook

Day	Date	Topic	Commit (on completion)
43	Mon, Jan 25, 2027	Design WhatsApp / Chat App	hld-day-43-whatsapp

Day	Date	Topic	Commit (on completion)
44	Tue, Jan 26, 2027	Design Instagram/Twitter Feed	hld-day-44-social-feed
45	Wed, Jan 27, 2027	Design Netflix / YouTube	hld-day-45-video-streaming
46	Thu, Jan 28, 2027	Design Notification & Scheduling System	hld-day-46-notification-job-scheduler
47	Fri, Jan 29, 2027	Design Uber — Part 1 (Capstone)	hld-day-47-uber-capstone-draft
48	Mon, Feb 01, 2027	Design Uber — Part 2, Capstone Finalize	hld-day-48-capstone-final

Reference Section — Full Repo List (all 15, for HLD phase)

1. design-guru/grokking-system-design
2. yangshun/tech-interview-handbook
3. kdn251/interviews
4. alex/what-happens-when
5. Chalarangelo/30-seconds-of-code
6. codersguild/System-Design
7. arpit20adlakha/Computer-Science-Papers-For-System-Design
8. InterviewReady/system-design-resources
9. arpit20adlakha/Data-Structure-Algorithms-LLD-HLD
10. kilimchoi/engineering-blogs
11. donnemartin/system-design-primer
12. addyosmani/essential-js-design-patterns
13. binhnguyennus/awesome-scalability
14. ashishps1/awesome-system-design-resources
15. CodingChallengesFYI/SharedSolutions

Tip: is file ko apne roadmap-tracker Streamlit app ke sath ya manually use kar sakte ho — jis din ka kaam ho jaye, us row ka commit message copy karke wahi commit kar dena.